

The most robust way to derive these formulas is using index notation (summation notation) combined with the definition of the partial derivative. This method avoids the confusion that often arises from trying to memorize vector/matrix rules directly.

Here is the step-by-step derivation for each formula.

Important Note on Layout Conventions

In matrix calculus, there are two common conventions. The results you provided ($\nabla_{\mathbf{x}} \mathbf{A}\mathbf{x} = \mathbf{A}^\top$) indicate you are using the Denominator Layout.

- **Numerator Layout (Jacobian):** The derivative of a vector $\mathbf{y}$ w.r.t (with respect to) $\mathbf{x}$ is a matrix where rows correspond to y and columns to x .
- **Denominator Layout (Gradient):** The derivative is the transpose of the Jacobian. The dimensions of the gradient match the dimensions of the variable we are differentiating with respect to (i.e., if $\mathbf{x}$ is a column vector, the gradient is a column vector).

The derivations below utilize the **Denominator Layout** to match your requested formulas.

The Linear Forms

Derive $\nabla_{\mathbf{x}} \mathbf{A}\mathbf{x} = \mathbf{A}^\top$

Let $\mathbf{y} = \mathbf{A}\mathbf{x}$. We want to find the derivative of the vector $\mathbf{y}$ with respect to vector $\mathbf{x}$.

1. Write in index notation: The i -th component of $\mathbf{y}$ is:

$$y_i = \sum_{k=1}^n A_{ik} x_k$$

2. Take the partial derivative: Differentiate y_i with respect to a specific component x_j :

$$\frac{\partial y_i}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\sum_k A_{ik} x_k \right)$$

Since x_j is the only variable that varies, the term is non-zero only when $k = j$.

$$\frac{\partial y_i}{\partial x_j} = A_{ij}$$

3. Assemble the matrix: In Denominator Layout, the derivative matrix is defined such that the (j, i) entry is $\frac{\partial y_i}{\partial x_j}$.

$$(\nabla_{\mathbf{x}} \mathbf{A} \mathbf{x})_{ji} = A_{ij}$$

Since the (j, i) entry is A_{ij} , the resulting matrix is the transpose of $\mathbf{A}$.

$$\nabla_{\mathbf{x}} \mathbf{A} \mathbf{x} = \mathbf{A}^{\top}$$

Derive $\nabla_{\mathbf{x}} \mathbf{x}^{\top} \mathbf{A} = \mathbf{A}$

Let $\mathbf{y}^{\top} = \mathbf{x}^{\top} \mathbf{A}$ (a row vector).

1. Write in index notation: The j -th component of the vector $\mathbf{y}^{\top}$ is:

$$(\mathbf{y}^{\top})_j = \sum_{k=1}^m x_k A_{kj}$$

2. Take the partial derivative: Differentiate the j -th component with respect to x_i :

$$\frac{\partial (\mathbf{y}^{\top})_j}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\sum_k x_k A_{kj} \right)$$

The derivative is non-zero only when $k = i$:

$$\frac{\partial (\mathbf{y}^{\top})_j}{\partial x_i} = A_{ij}$$

3. Assemble the matrix: Arranging these derivatives according to Denominator Layout results in the matrix $\mathbf{A}$ itself.

$$\nabla_{\mathbf{x}} \mathbf{x}^{\top} \mathbf{A} = \mathbf{A}$$

The Quadratic Form

Derive $\nabla_{\mathbf{x}} \mathbf{x}^{\top} \mathbf{A} \mathbf{x} = (\mathbf{A} + \mathbf{A}^{\top}) \mathbf{x}$

Let the scalar function be $f = \mathbf{x}^{\top} \mathbf{A} \mathbf{x}$.

1. Write in index notation:

$$f = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$$

2. Take the partial derivative with respect to x_k : We use the product rule because x_k appears twice (once as x_i and once as x_j).

$$\frac{\partial f}{\partial x_k} = \sum_i \sum_j A_{ij} \frac{\partial}{\partial x_k} (x_i x_j)$$

Using the product rule $\frac{\partial(x_i x_j)}{\partial x_k} = \frac{\partial x_i}{\partial x_k} x_j + x_i \frac{\partial x_j}{\partial x_k}$:

$\frac{\partial x_i}{\partial x_k}$ is 1 if $i = k$, else 0 (Kronecker delta δ_{ik}).

$\frac{\partial x_j}{\partial x_k}$ is 1 if $j = k$, else 0 (Kronecker delta δ_{jk}).

Substituting this back:

$$\frac{\partial f}{\partial x_k} = \sum_i \sum_j A_{ij} (\delta_{ik} x_j + x_i \delta_{jk})$$

3. Split the sums:

$$\frac{\partial f}{\partial x_k} = \underbrace{\sum_j A_{kj} x_j}_{\text{Sum 1}} + \underbrace{\sum_i x_i A_{ik}}_{\text{Sum 2}}$$

- Sum 1 is the definition of the k -th element of vector $\mathbf{Ax}$.
- Sum 2 is the definition of the k -th element of vector $\mathbf{A}^\top \mathbf{x}$ (since $\sum x_i A_{ik} = \sum (A^\top)_{ki} x_i$).

4. Combine into vector form: The gradient vector $\nabla_{\mathbf{x}} f$ is composed of these partials for all k :

$$\nabla_{\mathbf{x}} \mathbf{x}^\top \mathbf{Ax} = \mathbf{Ax} + \mathbf{A}^\top \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}$$

Special Case: $\|\mathbf{x}\|^2$

If $\mathbf{A} = \mathbf{I}$ (Identity matrix), then $\mathbf{x}^\top \mathbf{Ix} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2$.

$$\nabla_{\mathbf{x}} \|\mathbf{x}\|^2 = (\mathbf{I} + \mathbf{I}^\top) \mathbf{x} = (2\mathbf{I}) \mathbf{x} = 2\mathbf{x}$$

The Frobenius Norm

Derive $\nabla_{\mathbf{X}} \|\mathbf{X}\|_F^2 = 2\mathbf{X}$

The Frobenius norm squared is the sum of the squares of all elements in the matrix.

1. Write in index notation:

$$f(\mathbf{X}) = \|\mathbf{X}\|_F^2 = \sum_{i=1}^m \sum_{j=1}^n X_{ij}^2$$

2. Take the partial derivative: We want the gradient with respect to the matrix $\mathbf{X}$, which means we calculate the partial derivative with respect to a specific entry X_{kl} .

$$\frac{\partial f}{\partial X_{kl}} = \frac{\partial}{\partial X_{kl}} \left(\sum_i \sum_j X_{ij}^2 \right)$$

All terms in the sum where $(i, j) \neq (k, l)$ are treated as constants and their derivative is 0. We are left with only the derivative of the X_{kl}^2 term.

$$\frac{\partial f}{\partial X_{kl}} = \frac{\partial}{\partial X_{kl}} (X_{kl}^2) = 2X_{kl}$$

3. Assemble the matrix:

The gradient is a matrix where the (k, l) entry is $2X_{kl}$.

$$\nabla_{\mathbf{X}} \|\mathbf{X}\|_F^2 = 2\mathbf{X}$$

Summary for your reference

Expression	Derivative $\nabla \mathbf{f}$	Type
$\mathbf{A}\mathbf{x}$	$\mathbf{A}^\top$	Linear (Vector-valued)
$\mathbf{x}^\top \mathbf{A}$	$\mathbf{A}$	Linear (Vector-valued)
$\mathbf{x}^\top \mathbf{A}\mathbf{x}$	$(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$	Quadratic (Scalar)
$ \mathbf{x} $	$2\mathbf{x}$	Norm (Scalar)
$ \mathbf{X} _F$	$2\mathbf{X}$	Matrix Norm (Scalar)